

① 设 A 为 $m \times n$ 矩阵, 则 B 为 $n \times n$ 矩阵.

$$A^T A x = 0$$

$$x^T A^T A x = 0$$

$$(Ax)^T Ax = 0$$

则若 $A^T A x = 0$ 只有 $Ax = 0$

且若 $Ax = 0$ 则 $A^T(Ax) = A^T \cdot 0 = 0$

$$\therefore \text{Null } A = \text{Null } A^T A$$

$$\therefore \dim \text{Null } A = \dim \text{Null } A^T A \quad \text{即 } \dim \text{Null } B = \dim \text{Null } A$$

$$\therefore \text{rank } B = \text{rank } A$$

$$\textcircled{2} B = U_B D_B U_B^T$$

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$$

$$\lambda_{r+1} = \lambda_{r+2} = \dots = \lambda_n = 0$$

求证 $r = \text{rank } B$ 由谱分解易知 U_B 线性无关，
每个列向量为与 D_B 中特征值对应的特征向量 $\text{rank } U_B = r$
由题总可知 D_B 为对角矩阵，对角线为特征值

$$B = \sum_{i=1}^n d_i \mu_i \mu_i^T$$

可知 B 的列空间是由 $U_B \cdot U_B^T$ 的列空间的线性组合。

$$\therefore \text{rank } B \leq \text{rank } U_B \cdot U_B^T = \text{rank } U_B = r.$$

又，非零特征值对应的特征向量通过线性组合表示， $\therefore \text{rank } B \geq r$

$$\therefore \text{综上 } \text{rank } B = r$$